MARMA-CRYPTO PROTOCOL v1.2.1 · SOVEREIGN DOCUMENT SEALING

Cryptographic Geometric Watermarking via 54-Marma Sri Yantra & DSTU 4145-2002

Bind document text directly into the nonlinear geometry of a visible vector seal.



1. Theorem 1: Euclidean Impossibility Proof

54 triple intersections (marma-sthana) solved to IEEE-754 machine zero (6.13e-11 rad).

2. 24-Bit Quantized Visual Commitment B(K)

Maps 512-bit cryptographic digest into 4,096 nonlinearly coupled triangle states.

3. PAdES-B-LTA & CAdES-X-Long + RFC 3161 TSP

Full ETSI EN 319 142-1 archival LTV sealing over Galois field GF(2^257).

4. Instant Zero-Dependency Browser Verifier

Verify PDF/DOCX/SVG & simulate 1-bit tamper attacks live in the browser.

CERN Zenodo DOI: 10.5281/zenodo.23436152 | Apache-2.0 Open Source

yaroslavbaienko.github.io/marma-crypto-protocol

ZERO-TOLERANCE FORENSIC VERIFICATION · ATTACKS I–III

Why Detached PDF Signatures Fail
& How Geometric Commitments Survive

Detects 1-bit text edits, classical vector redrawing, and signature stripping.



Attack Class I: Post-Sealing Payload Swap

1-bit text modification shifts K' != K and breaks all 36 triangle coordinates.

Attack Class II: Vector Editor Forgery

Standard editors produce angular defect > 1e-2 rad vs threshold tau = 1e-9 rad.

Attack Class III: Signature Stripping / Print

Even if CMS/PKCS#7 wrapper is stripped, visible marma topology preserves K.

9-Suite Cross-Verification Harness

Verified in Python, Browser WebCrypto JS, and native Ukrainian QTSP CZO core.

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100% Detection Across All 36 Coordinates

DEBIAN / UBUNTU OFFICIAL APT REPOSITORY · GNOME NAUTILUS

1-Command Linux Installation

CLI + GTK4/Libadwaita + Nautilus Scripts

Seamless desktop integration for lawyers, researchers, and security engineers.



Official GPG-Signed APT Repository

sudo apt update && sudo apt install qes-tools marma-crypto-watermark

GNOME Files (Nautilus) Right-Click Menu

1-click stamping, verification, and official bilingual PDF Verification Protocols.

Hardware USB Token & Key Store Support

Works with ~/.secure_keys/ (.pfx/.p12/.key) and hardware tokens via GTK4 dialog.

Archival Academic Specification on CERN Zenodo

Bilingual EN/UA Whitepapers sealed with PAdES-B-LTA & RFC 3161 timestamp.

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yaroslavbaienko.github.io/legaltech

OPEN-CORE PROTOCOL (APACHE 2.0) · ENTERPRISE & FORENSIC ADVISORY

Enterprise Document Sealing & Legal Forensics

Bringing mathematical non-repudiation (Sri Yantra × DSTU 4145-2002 / PAdES-B-LTA) to high-stakes workflows:

PACKAGE 01 · CI/CD & API

Enterprise Document Sealing CI/CD

Automated integration of Marma-Crypto + Qualified Electronic Signatures (DSTU 4145-2002 / eIDAS PAdES-B-LTA + RFC 3161 TSP) into banking, M&A data rooms, law firms, and RWA/Web3 pipelines via Linux CLI, Docker, or REST

PACKAGE 02 · INSTITUTIONAL IDENTITY

Custom Corporate Geometric Seal

Dedicated organizational geometric-cryptographic seal bound to your corporate HSM / PKCS#11 / QES certificate, white-label browser verification portal, and brand-aligned vector templates for contracts & board resolutions.

PACKAGE 03 · LITIGATION & ARBITRATION

Forensic Authenticity & Expert Opinion

Independent cryptographic & geometric forensic audit of disputed electronic evidence (.pdf, .docx, .svg, .p7s), sub-micron tamper localization across all 36 triangle coordinates, and formal expert reports for courts &

 Verifier: yaroslavbaienko.github.io/marma-crypto-protocol | DOI: 10.5281/zenodo.23436152